

Lecture 1

Basics of Linear Algebra

- Matrix Operations
- Matrix Derivative and Expectations
- Applications and Wrap-Up

Contents

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- Matrix Derivative and Expectations
- Applications and Wrap-Up

Motivation

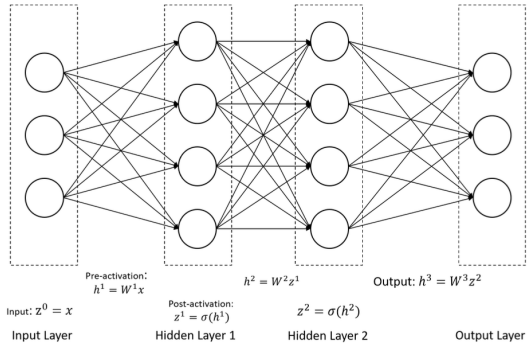


Figure: Example of a 3-layer fully-connected neural network. You should be able to understand its matrix representation.

What is a Matrix?

Let $A = (a_{ij})$ be an $m \times n$ matrix.

- The j th column of A is denoted by a column vector $\mathbf{a}_j$, i.e.,

$$\mathbf{a}_j = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

- The i th row of A is denoted by a row vector $\vec{\mathbf{a}}_i$, i.e.,

$$\vec{\mathbf{a}}_i = (a_{i1}, a_{i2}, \dots, a_{in})$$

- Matrix A can be represented in terms of either its columns and rows:

$$A = [\mathbf{a}_1, \dots, \mathbf{a}_n] = \begin{bmatrix} \vec{\mathbf{a}}_1 \\ \vec{\mathbf{a}}_2 \\ \vdots \\ \vec{\mathbf{a}}_m \end{bmatrix}$$

Matrix-Vector Multiplication

For an $m \times n$ matrix A with the i th column $\mathbf{a}_i$, and a vector $\mathbf{u} = (u_1, u_2, \dots, u_n)^\top$, the multiplication of A and $\mathbf{u}$ is defined as

$$A\mathbf{u} = u_1\mathbf{a}_1 + u_2\mathbf{a}_2 + \cdots + u_n\mathbf{a}_n$$

Example

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 6 \\ -7 \\ 8 \\ -9 \end{bmatrix} = 6 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - 7 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + 8 \begin{bmatrix} 3 \\ 4 \end{bmatrix} - 9 \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

Inner Product

- Given a vector $\mathbf{a} = (a_1, \dots, a_n)^\top$ and a vector $\mathbf{b} = (b_1, \dots, b_n)^\top$, following the rule of matrix-vector product, we have

$$\mathbf{a}^\top \mathbf{b} = a_1 b_1 + a_2 b_2 + \dots a_n b_n$$

- We call this special vector-vector multiplication the **inner product** (scalar product) of $\mathbf{a}$ and $\mathbf{b}$ (denoted by $\mathbf{a}^\top \mathbf{b}$ or $\langle \mathbf{a}, \mathbf{b} \rangle$)
- Properties: Commutative, bilinear
- Application: Cosine similarity, $\cos \theta = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$

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Row Perspective of Multiplication

The matrix-vector multiplication $A\mathbf{u}$ has a row formula as

$$A\mathbf{u} = \begin{bmatrix} \vec{\mathbf{a}}_1 \mathbf{u} \\ \vec{\mathbf{a}}_2 \mathbf{u} \\ \vdots \\ \vec{\mathbf{a}}_m \mathbf{u} \end{bmatrix}$$

- Consider $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} 6 & -7 & 8 & -9 \end{bmatrix}^\top$.
- We calculate

$$\vec{\mathbf{a}}_1 \mathbf{u} = 6 \cdot 1 - 7 \cdot 2 + 8 \cdot 3 - 9 \cdot 4 = -20$$

$$\vec{\mathbf{a}}_2 \mathbf{u} = 6 \cdot 2 - 7 \cdot 3 + 8 \cdot 4 - 9 \cdot 5 = -22$$

- We see that $A\mathbf{u} = \begin{bmatrix} -20 & -22 \end{bmatrix}^\top$

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Linear Systems as Matrix Equations

Write the following linear systems into compact matrix form:

$$\begin{cases} 2x_1 + x_2 + x_3 = 5 \\ 4x_1 - 6x_2 = -2 \\ -2x_1 + 7x_2 + 2x_3 = 9 \end{cases} \Rightarrow A\mathbf{x} = \mathbf{b}$$

where

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}$$

Rank of a Matrix

- The rank of a matrix A is the number of linearly independent columns
- Equivalently, it is the number of linearly independent rows
- Example: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has rank 1
- Full rank: $\text{rank}(A) = \min(m, n)$ for $A \in \mathbb{R}^{m \times n}$
- Application: Determines solvability of linear systems $A\mathbf{x} = \mathbf{b}$

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Identity Matrix

- The identity matrix of order k , denoted by I or I_k , is a $k \times k$ square matrix whose diagonal elements are 1's and whose nondiagonal elements are 0's

$$I = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

- Properties: $AI = A$ for any compatible matrix A .

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Inverse of a Matrix

- Let A be a $k \times k$ matrix. The inverse of A , denoted by A^{-1} , is another $k \times k$ matrix such that

$$AA^{-1} = A^{-1}A = I$$

- If the inverse exists, it is unique
- Existence: A^{-1} exists if and only if $\det(A) \neq 0$ (or equivalently, $\text{rank}(A) = k$)
- For 2×2 matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

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Transpose of a Matrix

- Let A be an $n \times k$ matrix. The transpose of A , denoted by $A^\top$, is a $k \times n$ matrix whose columns are the rows of A

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nk} \end{bmatrix} \Rightarrow A^\top = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & & \vdots \\ a_{1k} & a_{2k} & \cdots & a_{nk} \end{bmatrix}$$

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Symmetric Matrices

- Let A be a $k \times k$ matrix. A is said to be symmetric if

$$A = A^{\top}$$

- Examples: Covariance matrices, Hessian matrices
- Properties: Real eigenvalues, orthogonal eigenvectors
- Spectral theorem: $A = Q\Lambda Q^{\top}$ where Q is orthogonal and Λ is diagonal

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Idempotent Matrices

- Let A be a $k \times k$ matrix. A is called idempotent if

$$A = AA$$

- If A is also symmetric, then A is called symmetric idempotent
- If A is symmetric idempotent, then $I - A$ is also symmetric idempotent
- Example: Projection matrices $P = X(X^\top X)^{-1}X^\top$

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- Examples: Rotation matrices, permutation matrices

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Quadratic Forms

- Let $\mathbf{y}$ be a $k \times 1$ vector, and let A be a $k \times k$ matrix. The function

$$\mathbf{y}^\top A \mathbf{y} = \sum_{i=1}^k \sum_{j=1}^k a_{ij} y_i y_j$$

is called a quadratic form

- Geometric interpretation: Ellipsoids in k -dimensional space
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Positive Definite and Positive Semidefinite Matrices

Let A be a $k \times k$ matrix.

- A is said to be *positive definite* if

(a) $A = A^\top$ (A is symmetric)

(b) $\mathbf{y}^\top A \mathbf{y} > 0 \quad \forall \mathbf{y} \in \mathbb{R}^k, \mathbf{y} \neq 0$

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- Tests: Eigenvalues > 0 (positive definite), eigenvalues ≥ 0 (positive semidefinite)
- Application: Convex optimization, kernel methods

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Trace of a Matrix

Let A be a $k \times k$ matrix. The *trace* of A , denoted by $\text{trace}(A)$ or $\text{tr}(A)$, is the sum of the diagonal elements of A ; thus,

$$\text{trace}(A) = \sum_{i=1}^k a_{ii}$$

Properties:

1. If A is an $m \times n$ matrix and B is an $n \times m$ matrix, then

$$\text{trace}(AB) = \text{trace}(BA)$$

2. If the matrices are appropriately conformable, then

$$\text{trace}(ABC) = \text{trace}(CAB)$$

3. If A and B are $k \times k$ matrices and a and b are scalars, then

$$\text{trace}(aA + bB) = a\text{trace}(A) + b\text{trace}(B)$$

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Rank of an Idempotent Matrix

- Let A be an idempotent matrix. The rank of A is equal to its trace

$$\text{rank}(A) = \text{trace}(A)$$

- Proof sketch: Use the fact that idempotent matrices are diagonalizable with eigenvalues 0 or 1
- Application: In regression, $\text{rank}(X) = \text{trace}(H)$ where $H = X(X^\top X)^{-1}X^\top$ is the hat matrix

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An Important Identity for a Partitioned Matrix

Let $\mathbf{X}$ be an $n \times p$ matrix partitioned such that

$$\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$$

We note that

$$\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} = \mathbf{X}$$

$$\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top [\mathbf{X}_1 \ \mathbf{X}_2] = \mathbf{X}$$

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Consequently,

$$\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}_1 = \mathbf{X}_1 \quad \text{and} \quad \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}_2 = \mathbf{X}_2$$

Similarly,

$$\mathbf{X}_1^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{X}_1^\top \quad \text{and} \quad \mathbf{X}_2^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{X}_2^\top$$

Inverse of a Partitioned Matrix

Consider a matrix of the form

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} \mathbf{X}_1^\top \mathbf{X}_1 & \mathbf{X}_1^\top \mathbf{X}_2 \\ \mathbf{X}_2^\top \mathbf{X}_1 & \mathbf{X}_2^\top \mathbf{X}_2 \end{bmatrix}$$

It can be shown that the inverse of this matrix is $(\mathbf{X}^\top \mathbf{X})^{-1}$ that equals

$$\begin{bmatrix} (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} + (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2 G \mathbf{X}_2^\top \mathbf{X}_1 (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} & -(\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2 G \\ -G \mathbf{X}_2^\top \mathbf{X}_1 (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} & G \end{bmatrix}$$

where

$$\mathbf{H}_1 = \mathbf{X}_1 (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \quad \text{and} \quad G = [\mathbf{X}_2^\top (\mathbf{I} - \mathbf{H}_1) \mathbf{X}_2]^{-1}$$

Application: Regression analysis with multiple groups of predictors

Determinant

- The determinant of a square matrix A , denoted $\det(A)$ or $|A|$, is a scalar value
- Geometric interpretation: Scaling factor of the linear transformation
- For 2×2 matrix: $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$
- Properties:
 - $\det(AB) = \det(A) \det(B)$
 - $\det(A^{-1}) = 1 / \det(A)$
 - $\det(A^T) = \det(A)$
- Application: Testing invertibility, change of variables in integration

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- Application: Testing invertibility, change of variables in integration

Contents

- Matrix Operations
- Matrix Derivative and Expectations
- Applications and Wrap-Up

Matrix Derivatives

Let $\mathbf{A}$ be a $k \times k$ matrix of constants, $\mathbf{a}$ be a $k \times 1$ vector of constants, and $\mathbf{y}$ be a $k \times 1$ vector of variables.

1. If $z = \mathbf{a}^\top \mathbf{y}$, then

$$\frac{\partial z}{\partial \mathbf{y}} = \frac{\partial(\mathbf{a}^\top \mathbf{y})}{\partial \mathbf{y}} = \mathbf{a}$$

2. If $z = \mathbf{y}^\top \mathbf{y}$, then

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4. If $z = \mathbf{y}^\top \mathbf{A} \mathbf{y}$ and $\mathbf{A}$ is symmetric, then

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More Derivative Rules

- Application: Gradient descent optimization

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla f(\mathbf{w}_t)$$

where $\nabla f(\mathbf{w})$ is the gradient of the objective function

- Example: For linear regression with loss $L(\mathbf{w}) = \|\mathbf{y} - X\mathbf{w}\|^2$, the gradient is

$$\nabla L(\mathbf{w}) = -2X^\top(\mathbf{y} - X\mathbf{w})$$

- Chain rule for matrix derivatives: If $z = f(\mathbf{y})$ and $\mathbf{y} = g(\mathbf{x})$, then

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Expectations of Random Vectors

Let $\mathbf{A}$ be a $k \times k$ matrix of constants, $\mathbf{a}$ be a $k \times 1$ vector of constants, and $\mathbf{y}$ be a $k \times 1$ random vector with mean $\boldsymbol{\mu}$ and nonsingular variance–covariance matrix $\mathbf{V}$.

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4. $\text{Var}(\mathbf{A}\mathbf{y}) = \mathbf{A}\mathbf{V}\mathbf{A}^\top$

Note: If $\mathbf{V} = \sigma^2 \mathbf{I}$, then $\text{Var}(\mathbf{A}\mathbf{y}) = \sigma^2 \mathbf{A}\mathbf{A}^\top$

5. $\mathbb{E}(\mathbf{y}^\top \mathbf{A}\mathbf{y}) = \text{trace}(\mathbf{A}\mathbf{V}) + \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu}$

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Applications of Matrix Expectations

- Portfolio variance: For portfolio returns $\mathbf{r}$ with weights $\mathbf{w}$,

$$\text{Var}(\mathbf{w}^\top \mathbf{r}) = \mathbf{w}^\top \Sigma \mathbf{w}$$

where Σ is the covariance matrix of returns

- Risk estimation: For quadratic loss functions
- Signal processing: For estimating power in transformed signals
- Econometrics: In GMM and other estimation methods

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Applications in AI

- Neural networks: Weight matrices and activation functions

$$\mathbf{h}^{(l)} = f(W^{(l)}\mathbf{h}^{(l-1)} + \mathbf{b}^{(l)})$$

- Principal Component Analysis (PCA): Eigendecomposition of covariance matrix

$$\Sigma = Q\Lambda Q^{\top}$$

- Linear regression: Least squares solution

$$\hat{\beta} = (X^{\top}X)^{-1}X^{\top}y$$

- Support Vector Machines: Quadratic optimization with linear constraints

Back Propagation: Overview and Motivation

- Loss function: $F(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(f_{\theta}(x_i), y_i)$
- Goal: Minimize $F(\theta)$ using gradient descent

$$\theta(t+1) = \theta(t) - \alpha_t \nabla F(\theta(t))$$

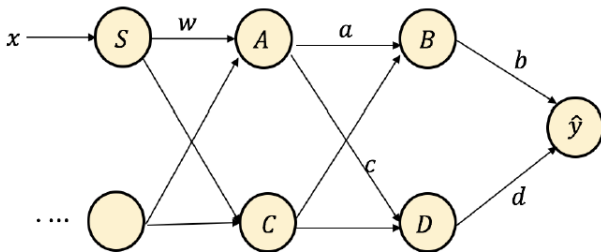
- Back propagation efficiently computes $\nabla F(\theta(t))$ using chain rule

Understanding BP in Level I: Scalar Form of Gradient

- Based on two fundamental rules:
 - Chain Rule: $\frac{df(g(w))}{dw} = \frac{df}{dg} \frac{dg}{dw}$
 - Sum Rule: $\frac{d(f_1(w)+f_2(w))}{dw} = \frac{df_1}{dw} + \frac{df_2}{dw}$
- Practical for coding implementations

Understanding BP in Level I: Scalar Form of Gradient

Example 2.2. Consider a 2-layer neural network with scalar output. We are interested in computing the derivative of this output $\hat{y}$ over a scalar parameter w . This function w.r.t. w can be represented in graph:



The computation of $\frac{\partial \hat{y}}{\partial w}$ can be summarized as follows:

BP Level II: Matrix Form Understanding

- Consider a 2-layer linear network (The weight matrices U, V are parameterized by θ) $f_{\theta}(x) = UVx$.
- Given n data points (x_i, y_i) , the goal is to minimize the loss function

$$F \triangleq \frac{1}{n} \sum_{i=1}^n \|UVx_i - y_i\|^2,$$

with U, V to be determined.

- The question is how to take gradient of F w.r.t. the matrix V ?
- Even simpler, how to compute $\frac{\partial F}{\partial V}$ with $F \triangleq \|UV - Y\|_F^2$? Here suppose that $U \in \mathbb{R}^{d_y \times d_1}$, $V \in \mathbb{R}^{d_1 \times d_x}$, $Y \in \mathbb{R}^{d_y \times d_x}$.

BP Level II: Matrix Form Understanding

- For $g(V) \triangleq \phi(Vx)$ with $x \in \mathbb{R}^{d \times 1}$ and $V \in \mathbb{R}^{k \times d}$, define $h = Vx$.

Then

$$\frac{\partial g}{\partial V} = \frac{\partial \phi}{\partial h} x^\top$$

- Exercise:

$$\frac{\partial \|AWB + C\|_F^2}{\partial W} = 2A^\top (AWB + C)B^\top$$

BP for General Deep Non-linear Network

Now derive the gradient of fully-connected neural network with quadratic loss. The objective f_{θ} is defined based on the following diagram:

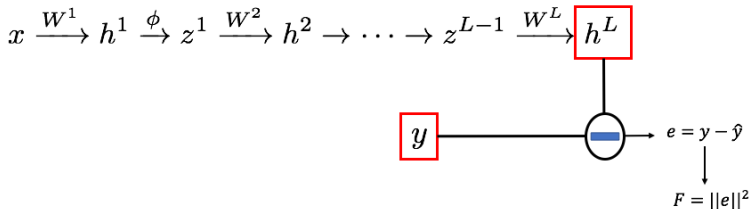


Figure: Diagram for the operator F

BP for General Deep Non-linear Network

- The derivative $\frac{\partial F}{\partial W^1}$ is computed as follows:

$$\frac{\partial F}{\partial W^1} = \frac{\partial F}{\partial h^1} x^\top \quad (1a)$$

$$= \left(\frac{\partial e}{\partial h^1} \right)^\top \left(\frac{\partial F}{\partial e} \right) x^\top \quad (1b)$$

$$= \left(\frac{\partial e}{\partial h^1} \right)^\top 2e \cdot x^\top$$
$$= \left(\frac{\partial e}{\partial h^L} \frac{\partial h^L}{\partial z^{L-1}} \cdots \frac{\partial h^1}{\partial z^1} \frac{\partial z^1}{\partial h^1} \right)^\top 2e \cdot x^\top \quad (1c)$$

$$= \left(W^L D^{L-1} W^{L-1} D^{L-2} \cdots W^2 D^1 \right)^\top 2e \cdot x^\top \quad (1d)$$

BP for General Deep Non-linear Network

- The general formula $\frac{\partial F}{\partial W^\ell}$ is left as exercise:

$$\frac{\partial F}{\partial W^\ell} = (W^L D^{L-1} \dots W^{\ell+1} D^\ell)^\top \cdot 2e \cdot (z^{\ell-1})^\top$$

This formula can be expressed in a recursive way, which is the mechanism of the BP technique.

- BP is an efficient way to compute all gradients $\frac{\partial F}{\partial W^\ell}$ for $\ell = 1, \dots, L$.
The naive computation complexity is $\mathcal{O}(d^2 L^2)$; while the BP complexity is $\mathcal{O}(d^2 L)$.

Further Reading

- Strang, G. (2016). *Introduction to Linear Algebra*
- Boyd, S. & Vandenberghe, L. (2018). *Introduction to Applied Linear Algebra*
- MIT OpenCourseWare: Linear Algebra